Business Recommendation Summary

Purchase Prediction Model — E-Commerce Mini-Project 5
Recommendations derived from model results and feature-importance analysis

1. Prioritize high-likelihood customers for remarketing

Model finding	Customers with high cart activity and prior purchase history score highest on predicted purchase probability.
Business interpretation	These customers are closest to converting and are the most cost-efficient marketing targets.
Recommended action	Serve personalized product recommendations, cart-abandonment reminders, and limited-time discounts; prioritize this segment in paid remarketing.
Expected benefit	Higher conversion rate per marketing dollar spent.
Risk / limitation	Over-discounting this group can erode margin on customers who may have converted without an incentive.

2. Nurture medium-likelihood customers with lower-cost tactics

Model finding	Customers with moderate engagement (time on site, pages viewed) but limited purchase history fall in the medium-probability band.
Business interpretation	These customers are interested but not yet convinced; more information or trust-building may close the gap.
Recommended action	Surface reviews and product comparisons, offer small incentives, send educational email content, and highlight free shipping/returns.
Expected benefit	Nudges borderline customers toward conversion without heavy discounting.
Risk / limitation	Response may be slower to materialize and harder to attribute to a single campaign.

3. Limit spend on low-likelihood customers

Model finding	Customers with low engagement and no discount/email/ad interaction show the lowest predicted purchase probability.

Business interpretation	This segment is not yet in a buying mindset; expensive campaigns are unlikely to pay off.
Recommended action	Use low-cost awareness content, collect feedback on drop-off points, improve product discovery and site usability, and retarget only once stronger engagement signals appear.
Expected benefit	Avoids wasted spend on unlikely buyers, freeing budget for higher-likelihood segments.
Risk / limitation	Some customers in this group may still convert later; suppressing outreach entirely risks losing them to a competitor.

4. Use on-site engagement as a real-time intent signal

Model finding	Cart items and time on site are the strongest positive predictors of purchase.
Business interpretation	On-site engagement is a real-time indicator of purchase intent, not just a historical one.
Recommended action	Trigger contextual nudges (chat prompts, urgency messaging) once a visitor crosses an engagement threshold within a session.
Expected benefit	Captures purchase intent in the moment rather than only after the visitor leaves the site.
Risk / limitation	Overly aggressive in-session prompts can feel intrusive and hurt the customer experience.

5. Automate recency-based re-engagement

Model finding	Purchase likelihood declines as the number of days since the last visit increases.
Business interpretation	Customer interest decays over time without contact.
Recommended action	Automate a re-engagement email/SMS sequence once a customer passes the recency threshold identified in the exploratory analysis.
Expected benefit	Recovers customers before they fully disengage from the brand.

Risk / limitation	Needs frequency capping to avoid contributing to unsubscribe or opt-out rates.